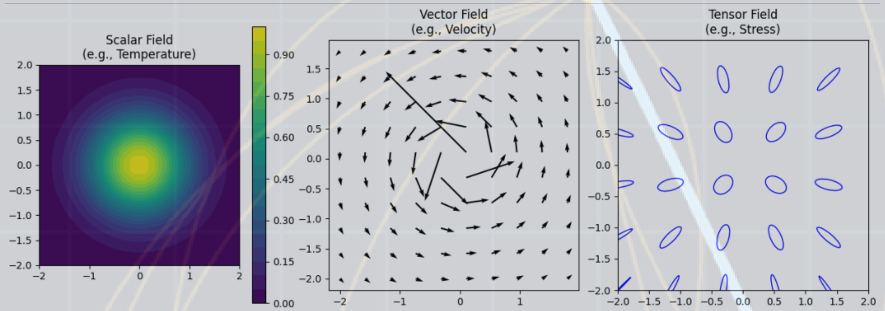

ENR-275 Classical Mechanics

Xiang Li

Fall 2026

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Portrait and field source: Wikipedia
Art-style change done by google nano banana 2

Course Syllabus and Discussion

In classical physics, if you know everything about a system at some instant of time, and you also know the equations that govern how the system changes, then you can predict the future.

- Leonard Susskind *Classical mechanics the theoretical minimum*

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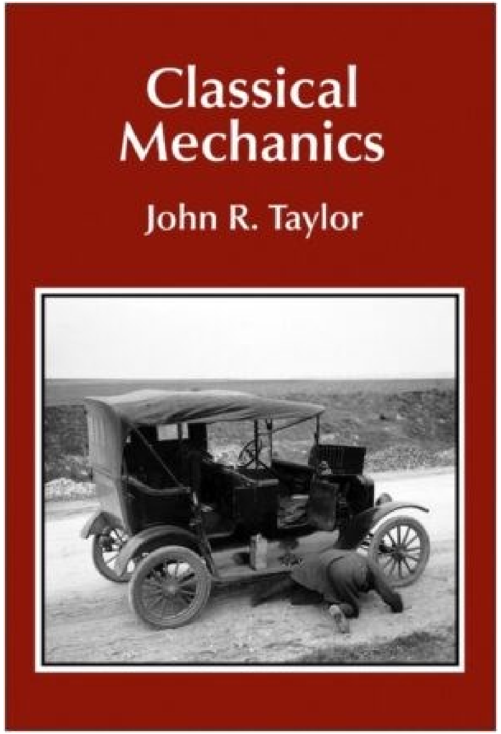
Course Syllabus and Discussion

Welcome to “Classic Mechs”:

- Pre-quantum
- Differential equations
- Linear algebra
- Little bit of iterative questions

Pre-requisition skills:

- ❑ Algebraic calculation and trigonometry
- ❑ Recall differential equations and vector calculus



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What is classic mech to me:

Modern "EVERYTHING"

What's next next next:

- Modern physics
- Modern Robotics
- AI

What's next next:

- Continuum mechanics
- Control systems
- Data science and machine learning

What's next:

- Complex analysis
- Vector calculus
- Fourier and Laplace transforms
- PDE

Least Action Principle, Lagrangian function, and Hamilton's equations

Calculus, ODE, Systems of ODE, Eigenvalues and Eigenvectors
Non-linear systems, Numerical and computational method

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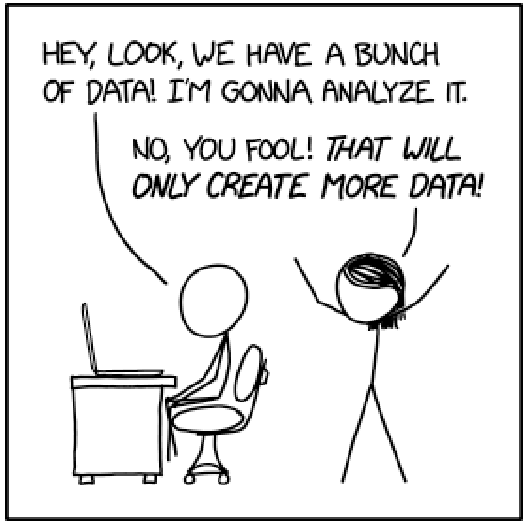

https://xkcd.com/2582/

Classic Mechs EP1: Analytical and probabilistic modeling

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Classic Mechs EP1: Analytical and probabilistic modeling

“Pre-lab”:

- Review Newton’s law
- Review Cartesian coordinates
- Review sliding blocks (1.36), will be in the quiz on Friday

Analytical modeling (1/2)

Modeling → Analytical

First Principle	(1.10)
Power law	part of (4.2)
Chain rule	to get (1.47)

Analytical modeling (2/2)

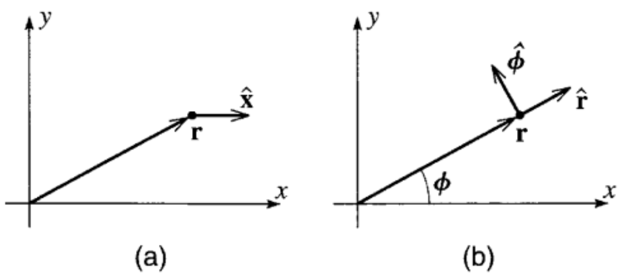

Figure 1.11 **(a)** The unit vector $\hat{\mathbf{x}}$ points in the direction of increasing x with y fixed. **(b)** The unit vector $\hat{\mathbf{r}}$ points in the direction of increasing r with ϕ fixed; $\hat{\boldsymbol{\phi}}$ points in the direction of increasing ϕ with r fixed. Unlike $\hat{\mathbf{x}}$, the vectors $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\phi}}$ change as the position vector $\mathbf{r}$ moves.

$$\begin{aligned}\Delta\hat{\mathbf{r}} &\approx \Delta\phi\,\hat{\boldsymbol{\phi}} \\ &\approx \dot{\phi}\,\Delta t\,\hat{\boldsymbol{\phi}}.\end{aligned} \qquad (1.41)$$

$$\frac{d\hat{\mathbf{r}}}{dt} = \dot{\phi}\,\hat{\boldsymbol{\phi}}. \qquad (1.42)$$

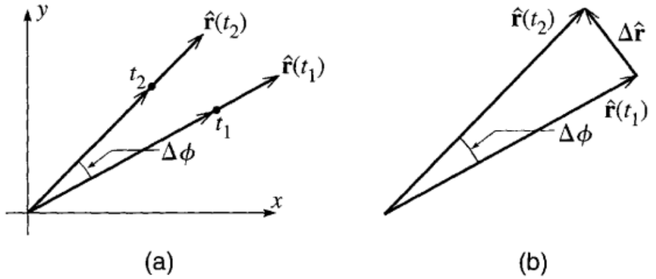

Figure 1.12 **(a)** The positions of a particle at two successive times, t_1 and t_2. Unless the particle is moving exactly radially, the corresponding unit vectors $\hat{\mathbf{r}}(t_1)$ and $\hat{\mathbf{r}}(t_2)$ point in different directions. **(b)** The change $\Delta\hat{\mathbf{r}}$ in $\hat{\mathbf{r}}$ is given by the triangle shown.

$$\frac{d\hat{\boldsymbol{\phi}}}{dt} = -\dot{\phi}\,\hat{\mathbf{r}}. \qquad (1.46)$$

$\hat{\phi}$ and $\overrightarrow{\phi}$ diff!

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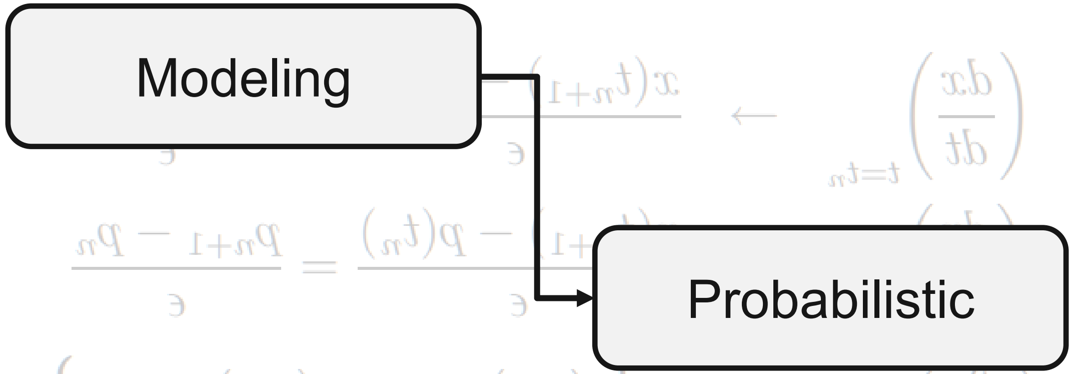
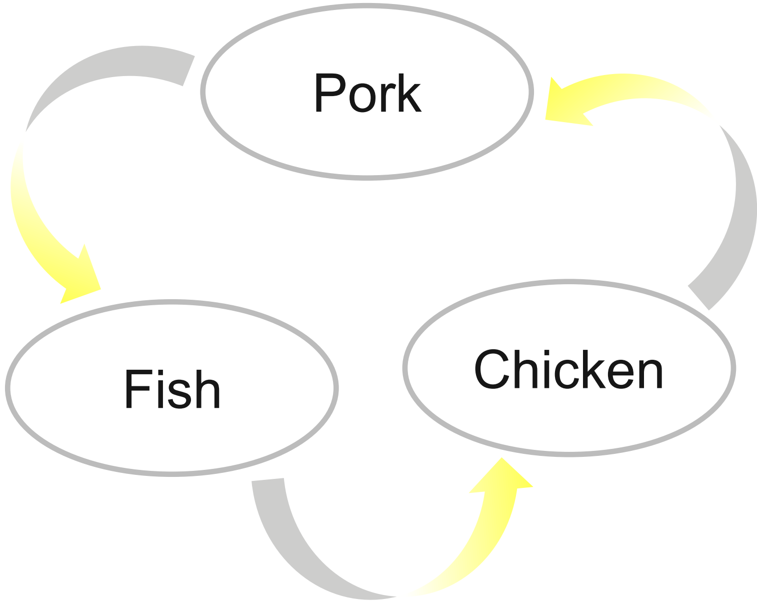
Probabilistic modeling, an example

I noticed that Simple Servings at the U is doing a rotation of proteins: Pork, Fish and Chicken

Can we do a **prediction** model?





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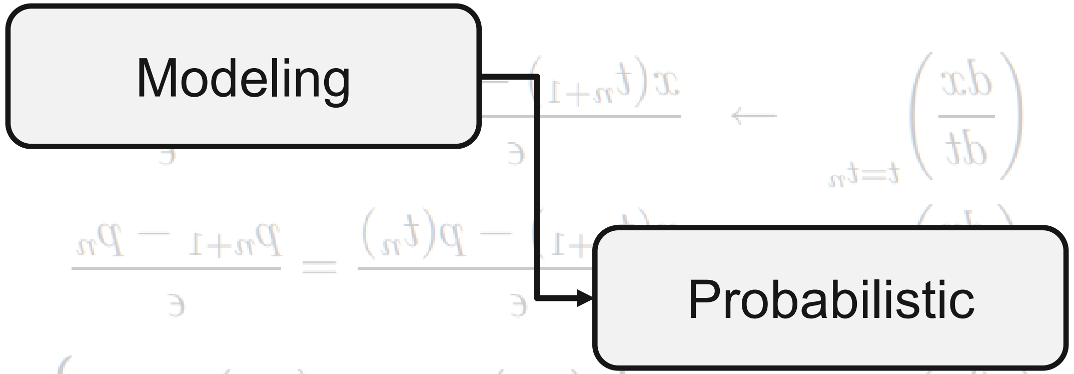
Probabilistic modeling, an example



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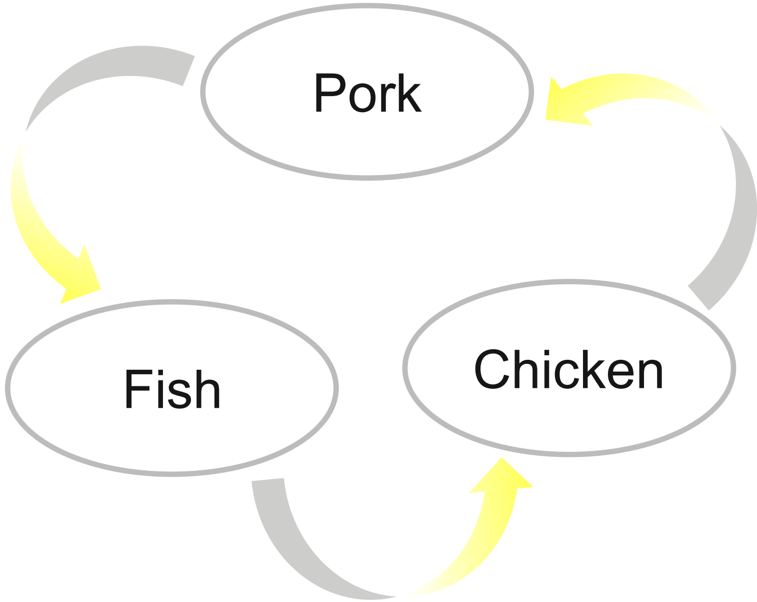
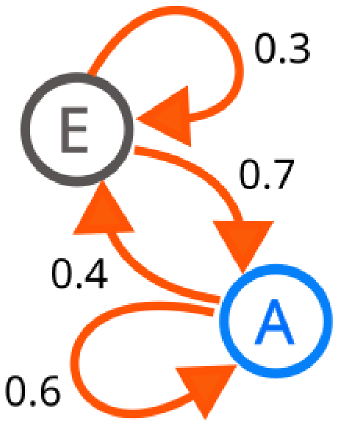


Probabilistic modeling, an example

Step 1: Markov chain



https://en.wikipedia.org/wiki/Markov_chain

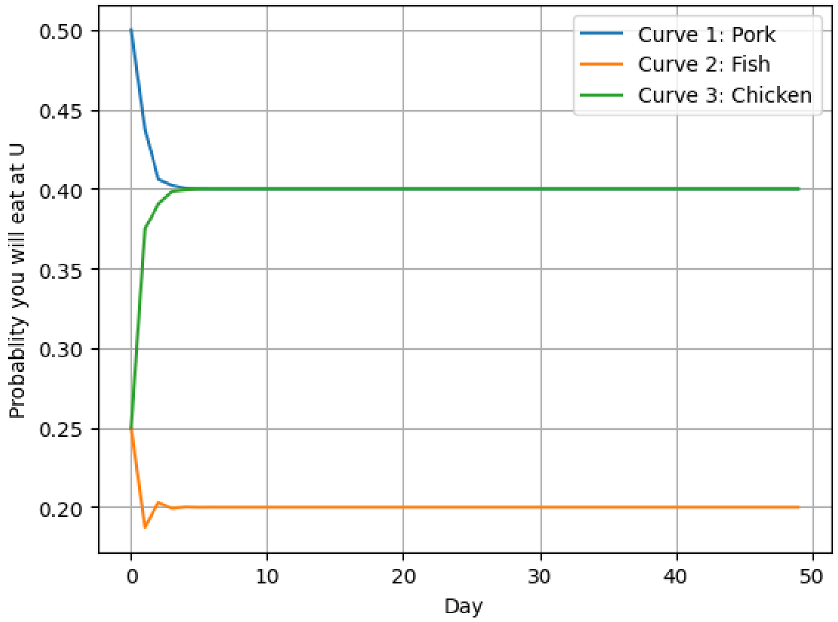
Step 2: Iteration in steps… in Python

```
A = np.array([[0.5, 0.5, 0.25], [0.25, 0, 0.25], [0.25, 0.5, 0.5]])
print (A)
```

```
[[0.5  0.5  0.25]
 [0.25 0.   0.25]
 [0.25 0.5  0.5 ]]
```

```
xtoday = [1, 0, 0]
```

```
for i in range(50):
    xtomorrow = A @ xtoday  # xtomorrow is linear combination of xtoday
    xtoday = xtomorrow
    the_food[i, :] = xtomorrow
```



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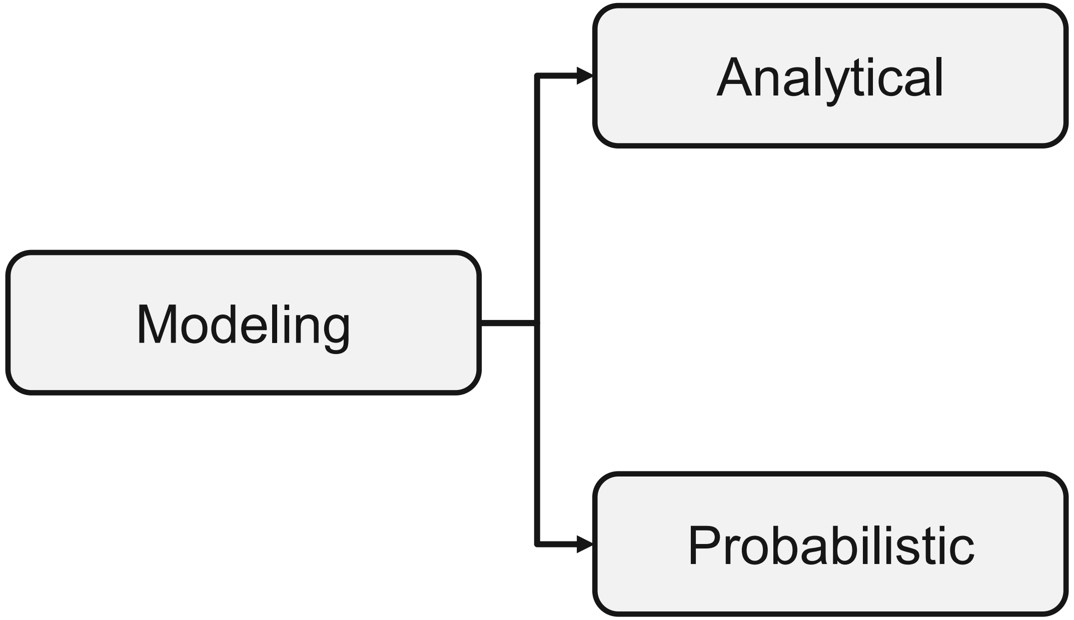
Episode review and homework (?)



First Principle	(1.10)
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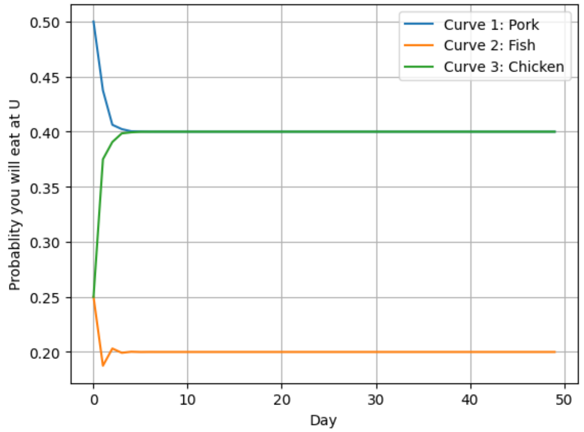

This is something off Taylor's textbook, but we will have Google Colab notebook to guide you through.

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